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ECG Introduction

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

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Session Objectives

1. Associate leads with their anatomical position and arteries
2. Identify and interpret orientation, rate, rhythm, axis, and intervals on an ECG
3. Recognize normal and abnormal waves, intervals, and segments.

First Aid (2025): 288, 292, 298, 308,310

Source: Course Syllabus

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The eyes cannot see what the mind does not know.

D.H. Lawrence

Wiggers Diagram

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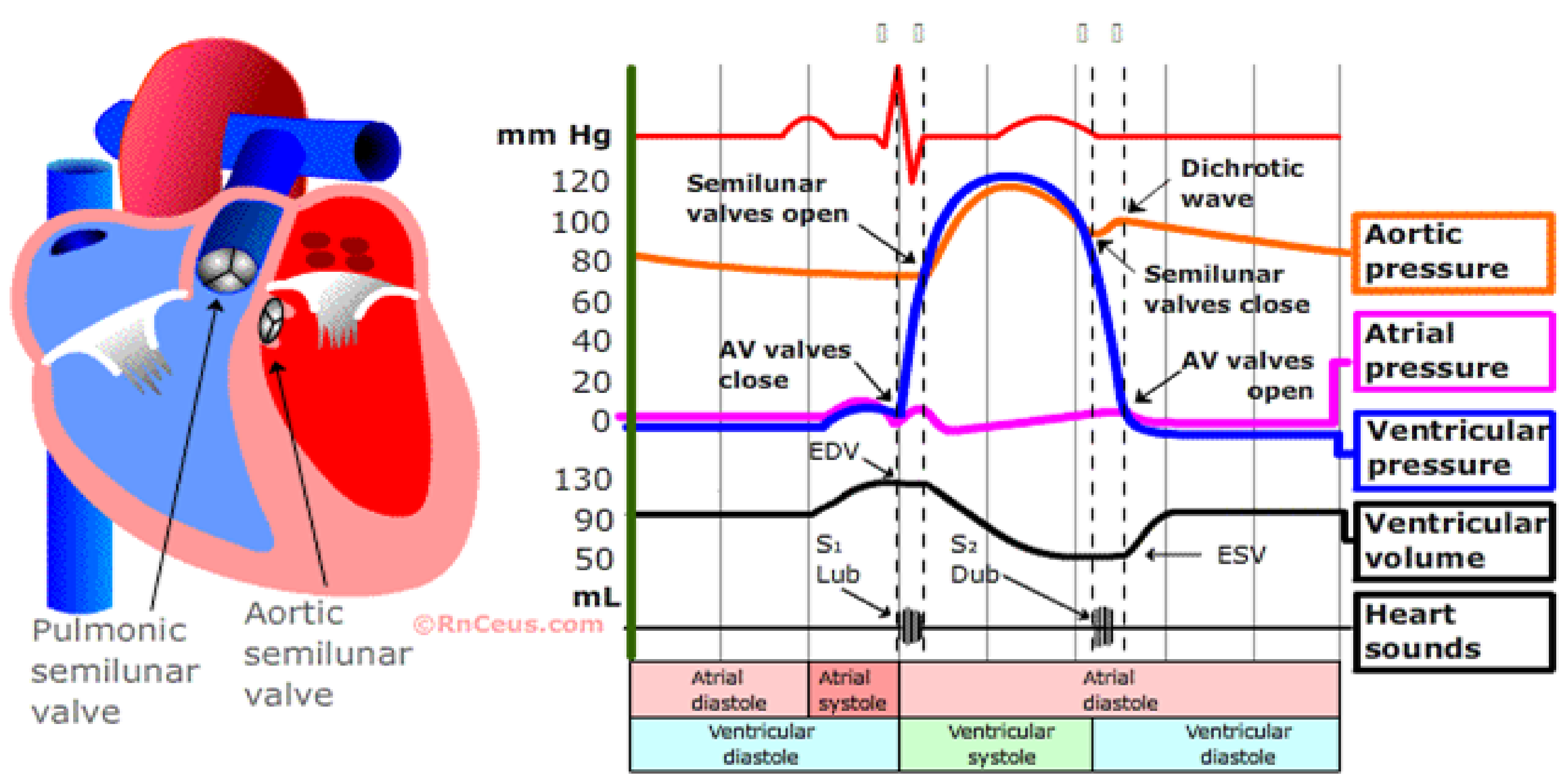

Coronary Circulation

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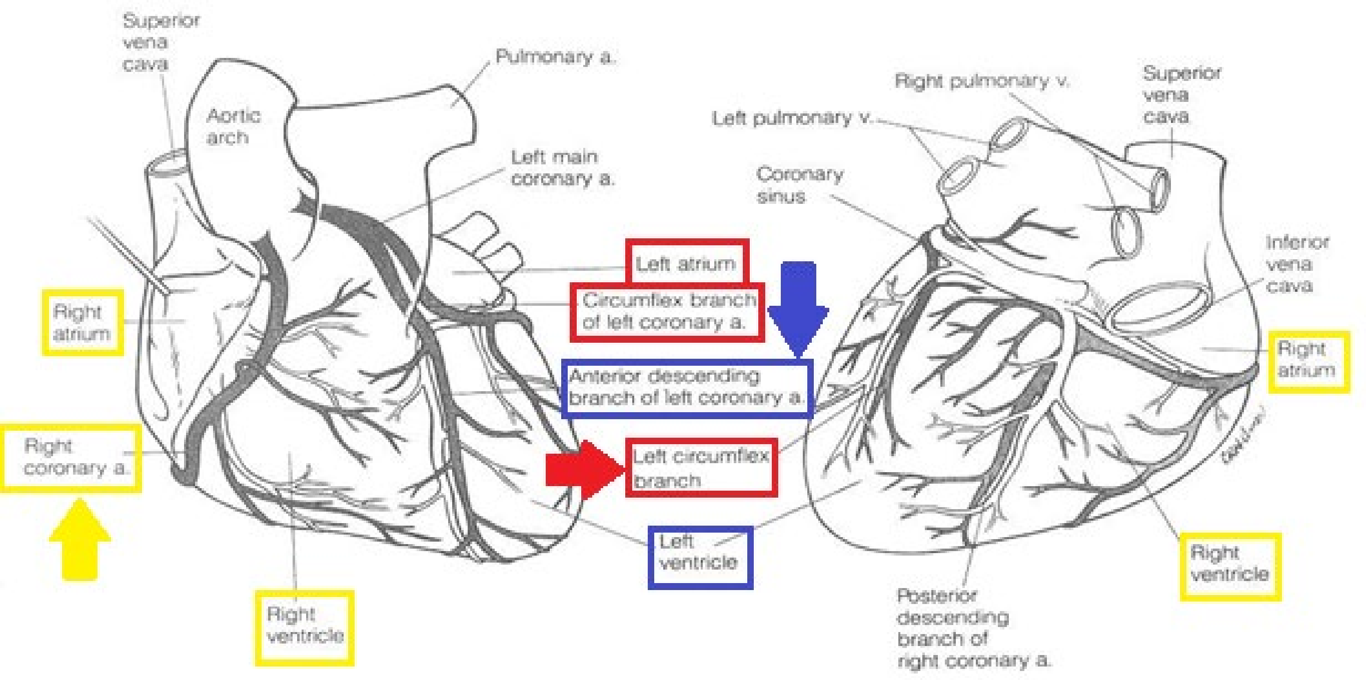
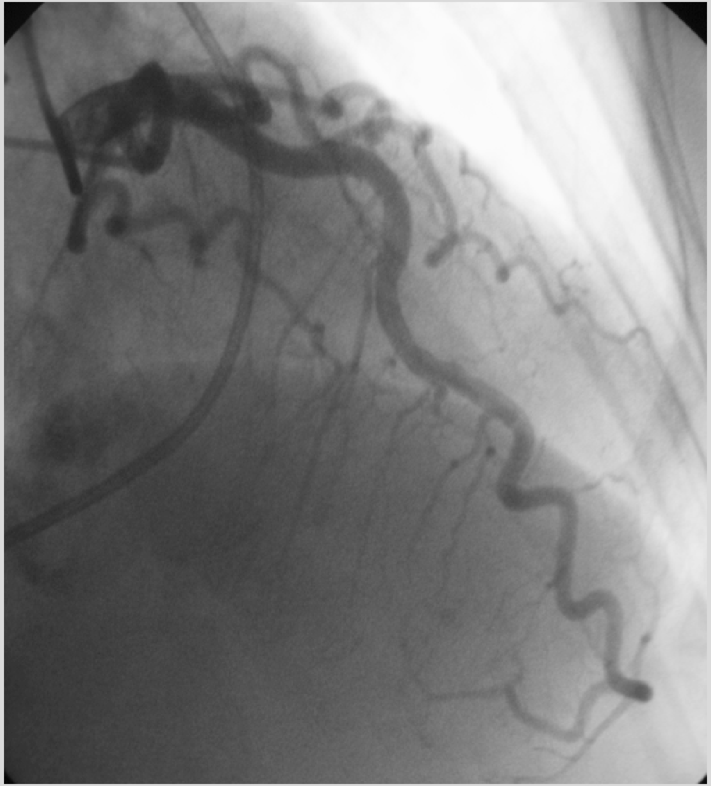


Left Circumflex Coronary Artery
"Lateral view"

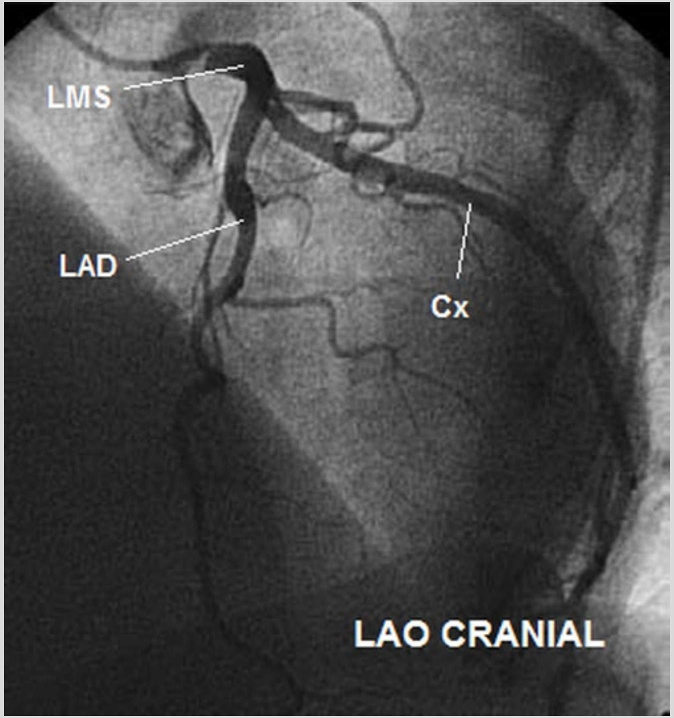
Left Anterior Descending Coronary Artery
"Septal & Anterior views"

Right Coronary Artery
"inferior view"

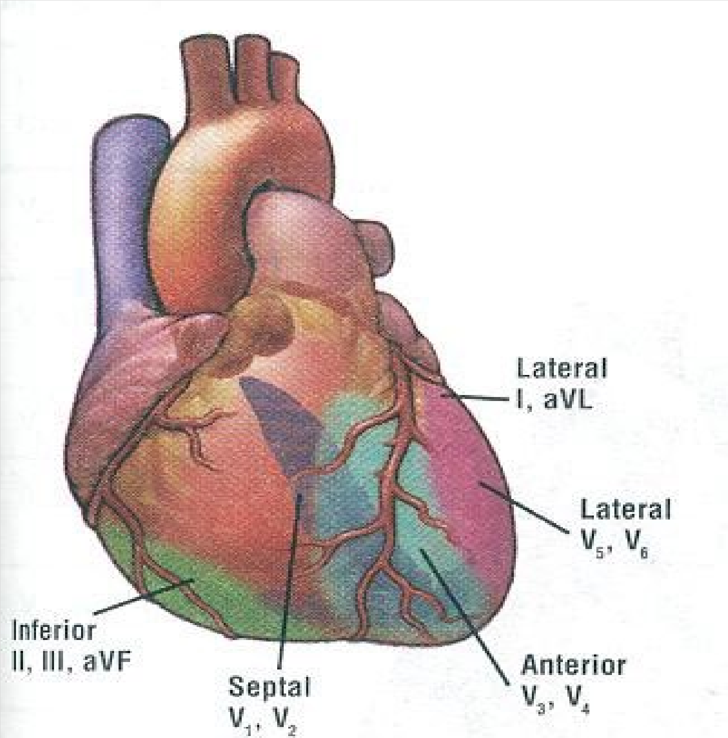
Angiography



LMS stands for “left main stem”



Corresponding Leads*



Right Coronary Artery (RCA):

- **Inferior** View
- **II,III, aVF**

Left Anterior Descending Artery (LAD):

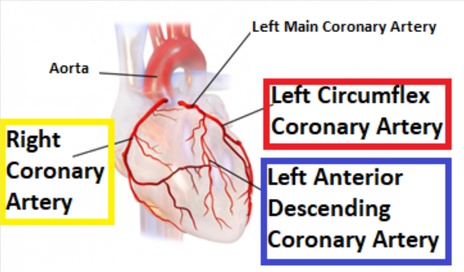
- **Septal** and **Anterior** Views
- **V1-V4**

Left Circumflex Artery (LCX):

- **Lateral** View
- **V5, V6, aVL**

*Reference Slide Notes

Corresponding Leads, Arteries, and Views



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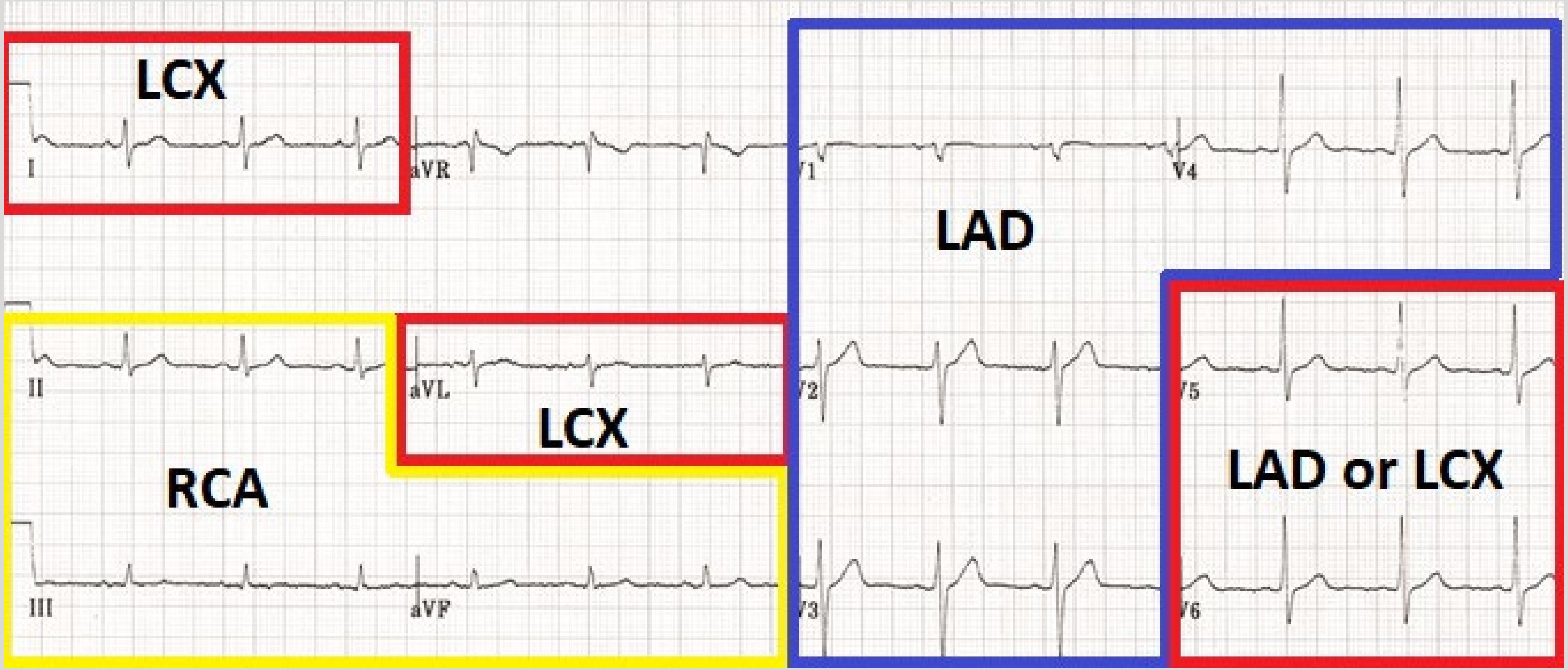

Mnemonic for Views: "SALI"

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S= septal; V1-V2 (LAD)

A= anterior; V3-V4 (LAD)

L= lateral; V5-V6, I, aVL (LCX/LAD)

I= inferior; II, III, aVF (RCA)

3D Anatomical View of Leads

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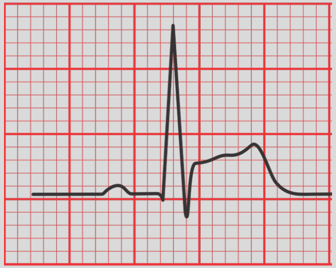
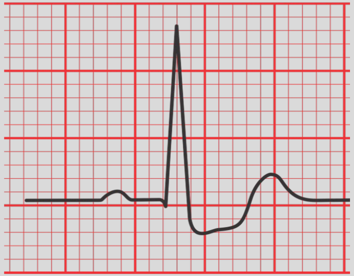
- https://skfb.ly/6QTXp

ST Segment

- ST segment “should” always be isoelectric
- <u>TP segment is the **true** isoelectric line</u>
- Elevation or depression represents either ischemia or injury
- ST elevation indicates transmural ischemia
- ST depression indicates subendocardial ischemia



ST elevation



ST Depression

https://lightssirensactionems.com/blogs/ems -medications/pails-ems-cardiology
https://www.cvphysiology.com/CAD/CAD012

Where is the STEMI?
(Clue: most common location)

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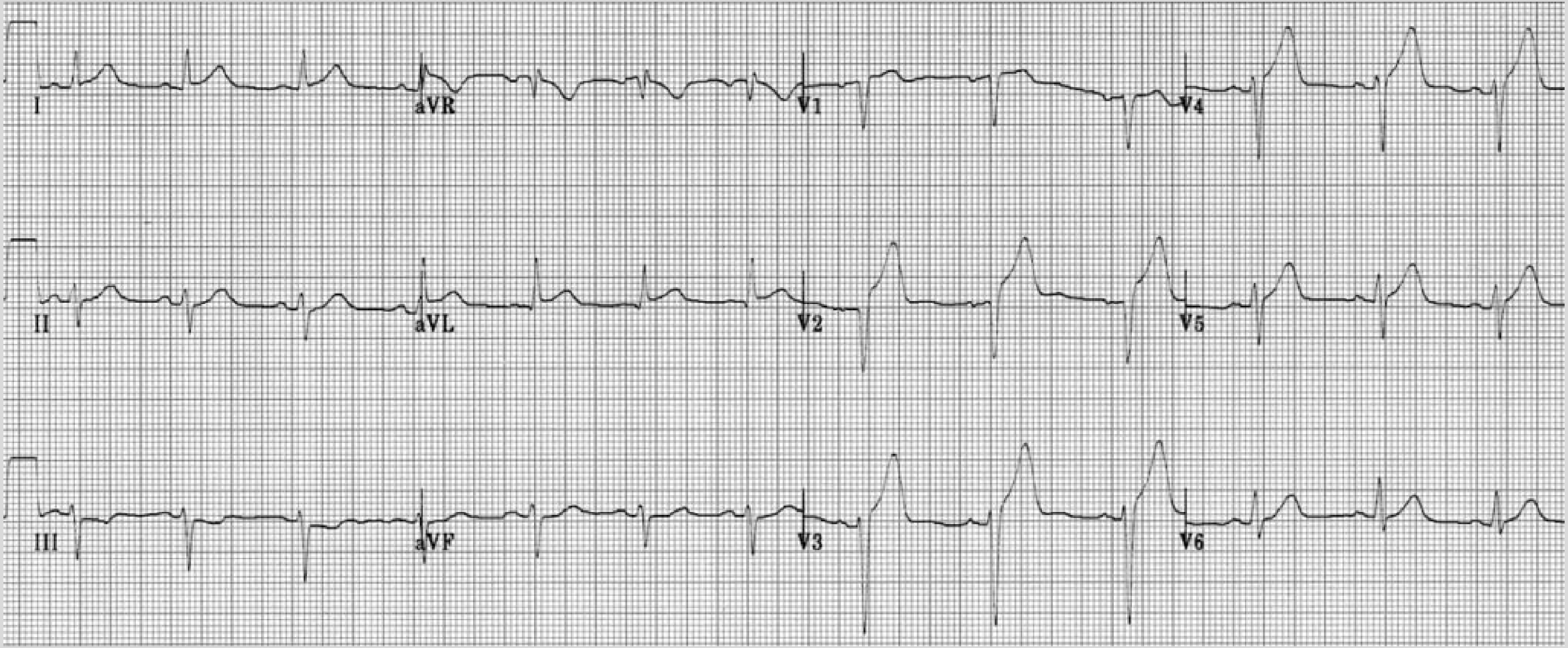

Where is the STEMI?

(Clue: 2nd most common location)

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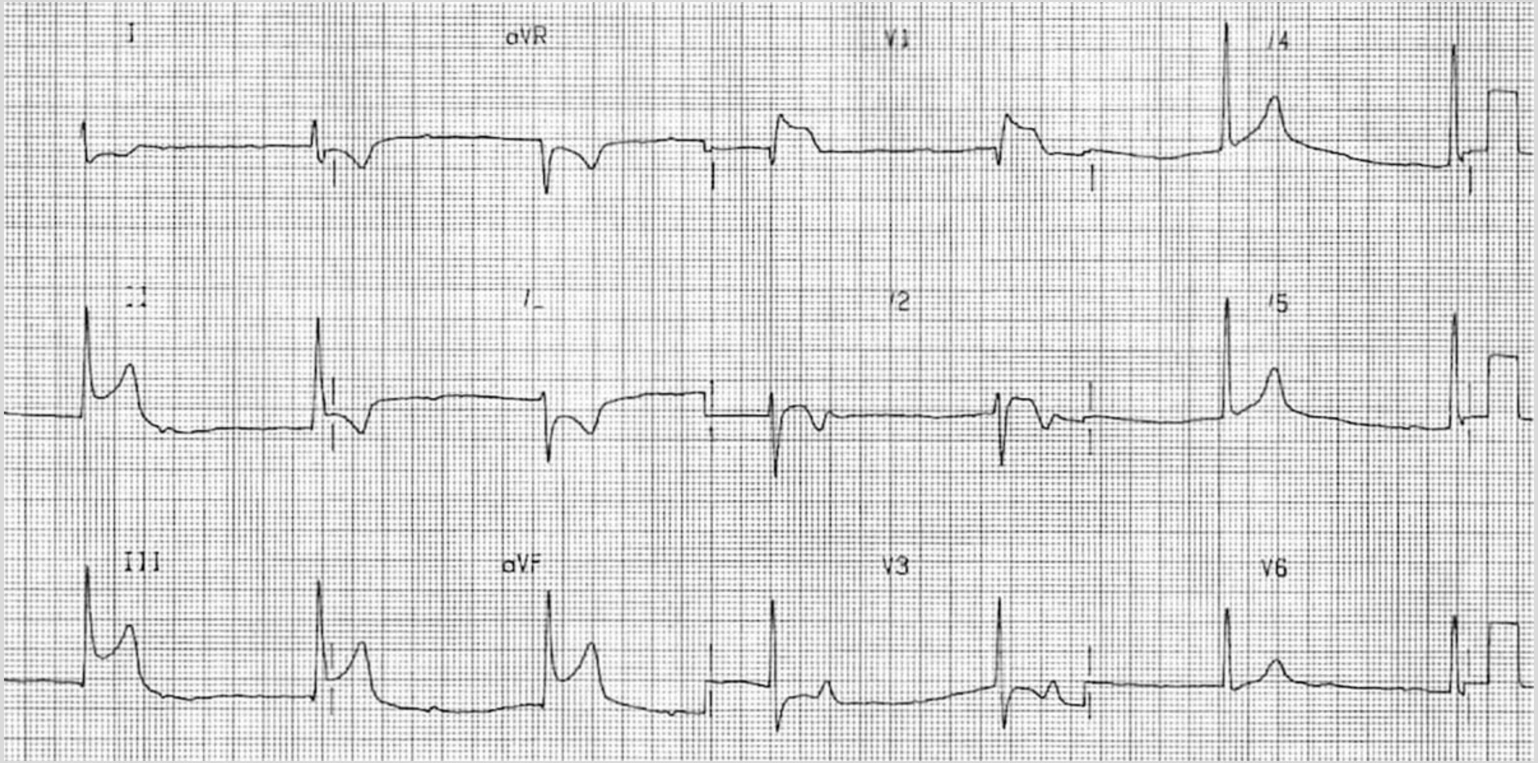

Objective 1 Questions

What of the following is not a correct correlation of cardiac view with leads?

LAD = V1-V4

RCA = II, III, aVF

LCX = V5-V6, I, aVL

none of the above

Objective 1 Questions

Pt presents with classic signs & sxs of an MI. On EKG, you find ST elevations in I & II. What do you diagnose?

Anterior STEMI

Inferior STEMI

Septal STEMI

Need more info to diagnose

Steps in Analyzing a 12-lead ECG

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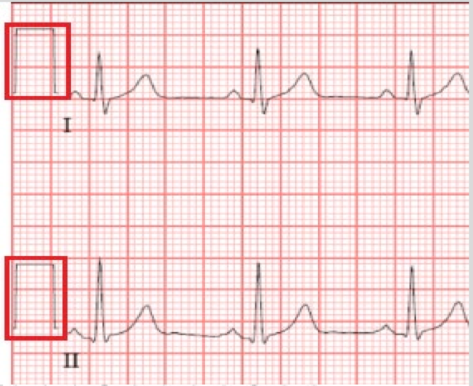
1. **Orient** yourself
2. Calculate **Rate**
3. Determine **Rhythm**
4. Determine the **Axis**
5. **Intervals** and QRS Duration
6. **"Others"** ST segment & T wave

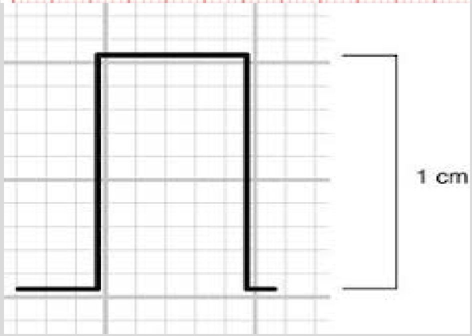
Officer **R**yan **R**eynolds **A**lways **I**mpresses **O**thers

Reading an ECG is a developed skill! Find the steps that work best for you and apply them in the same order for every ECG you read.

Orient Yourself

- Identify if this is a 12-lead or a rhythm strip
- Telemetry monitoring strips look very similar
- Check if the calibration is correct
  - Should be 2 boxes up and 1 box wide
- If you saw the patient note age, sex, body habitus
  - Thinner patients can have larger amplitudes
  - Obese patients can have smaller amplitudes





ECG Paper

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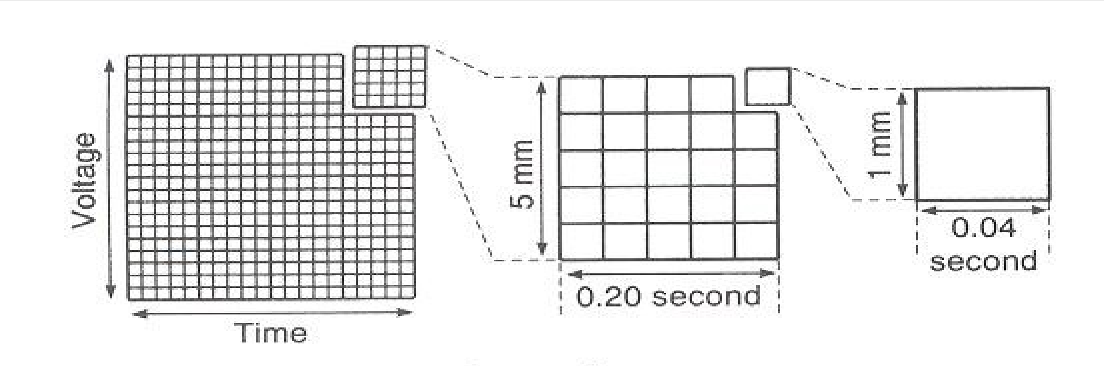


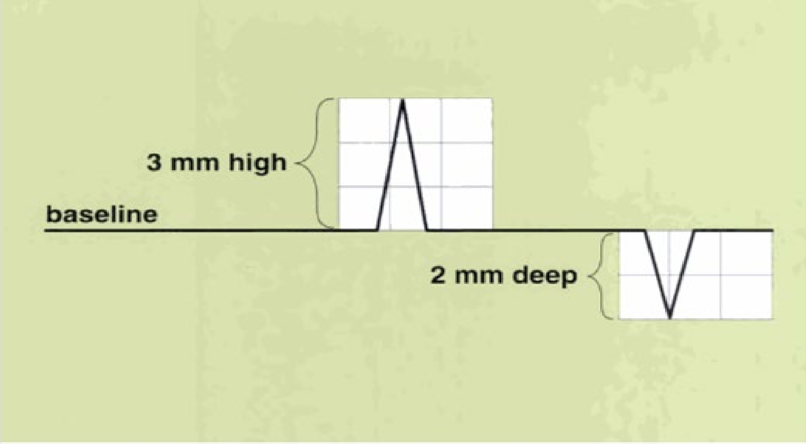
Horizontal axis = time

Vertical axis = voltage

1 small box = 0.04 seconds

1 large box = 5 small boxes

1 large box = 0.2 seconds



Dubin D. Rapid Interpretation of EKG's

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Calculate Rate

- Method 1: (# of QRS in entire strip X 6)
  - ECG strip represents 10 seconds X 6 = 60 seconds
- Method 2: Reference a QRS complex located on a vertical line. Every large square until the next QRS will provide estimation of the rate
  - Begins at 300 (1 square) then 150, 100, 75, 60, 50 for subsequent squares

Calculate Rate (ct’d)

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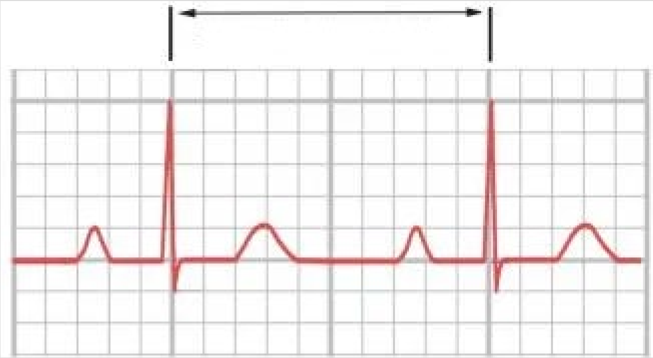
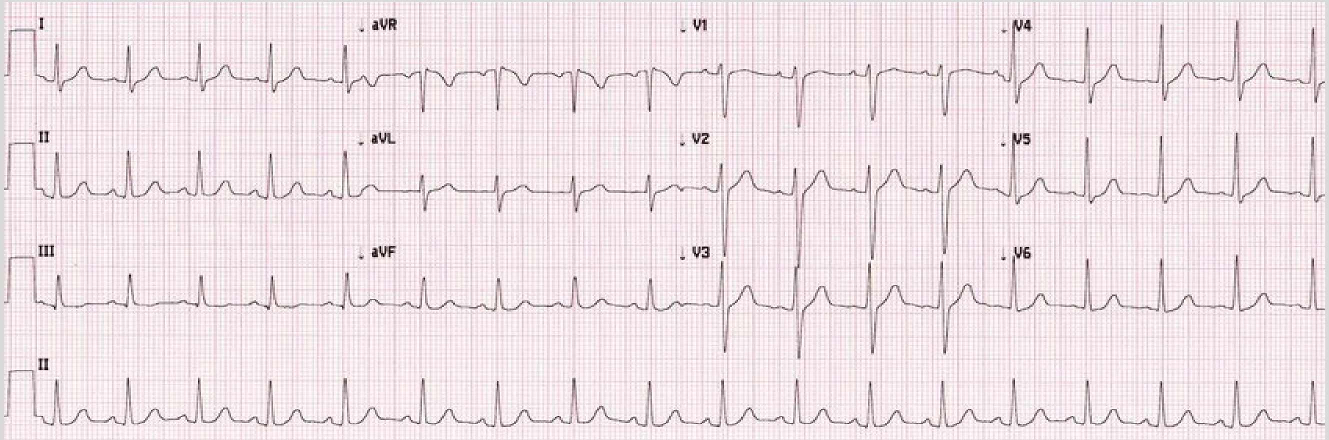
- Method 3: Count the number of big boxes between two QRS complexes and divide 300 by that number
  - Or 1500 divided by the number of small boxes
- Image below: 2 large boxes or 10 small boxes
  - 300/2 or 1500/10 gives a heart rate of 150 bpm



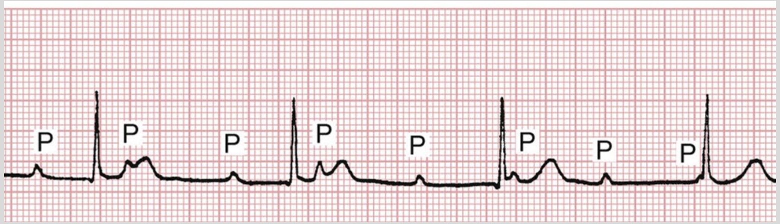
Calculate Rate (ct’d)

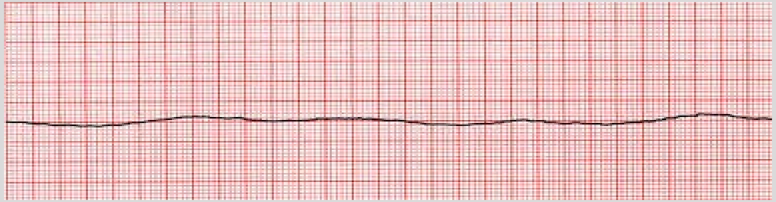


- Method 1: 18 QRS Complexes X 6 = 108
- Method 2: 3 squares in between QRS complexes (300 → 150 → 100) approximately 100 bpm
- Method 3: 300/(2.9 large boxes) = 103 bpm
  - 1500/(14 small boxes) = 108 bpm

Determine Rhythm

- Look for P waves (best leads are V1, II, III, aVF)
- Does every P wave precede a QRS complex?
- Are all P waves identical in size, shape, and position?





https://ekg.academy/learn -ekg?courseid

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Determine Rhythm (ct’d)

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- Next step is to see if the rhythm is regular or irregular
- Easiest method is to take a separate piece of paper and mark the R wave on 3-4 QRS complexes
- Next slide this down the ECG strip
- If the distance between R waves remains the same the rhythm is regular
- If the distance changes between R waves on the strip the rhythm is irregular

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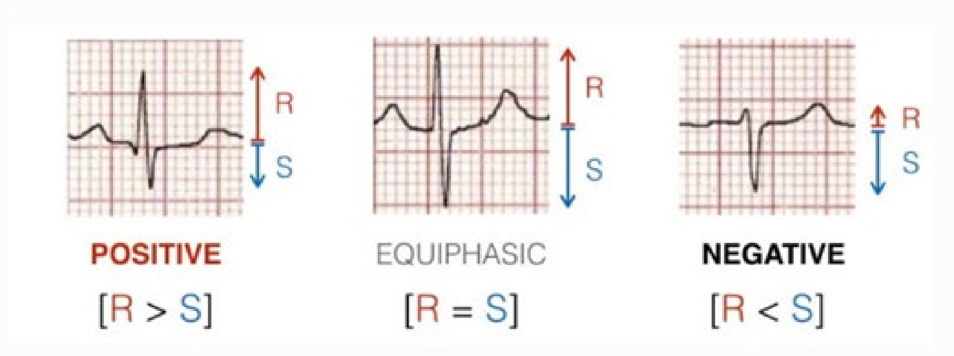
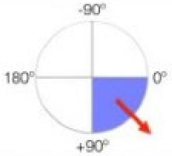
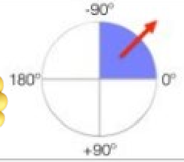
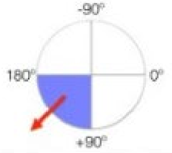
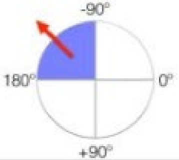
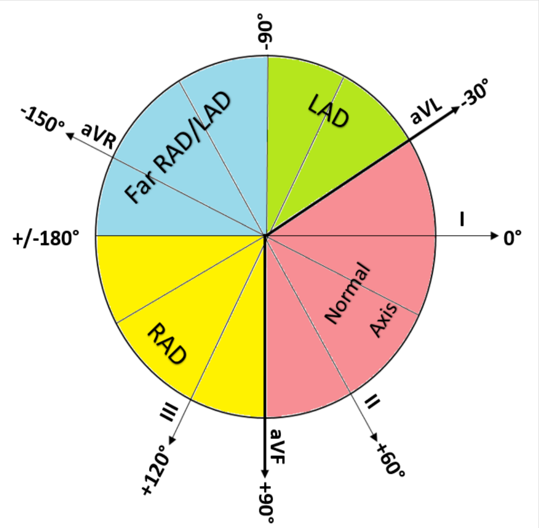
Determine the Axis

- The axis represents the overall direction of the depolarization of the heart
- Most efficient way to estimate the axis is look at the QRS complex in Lead I and Lead aVF
- Identify if the QRS complex is positive, negative, or equiphasic



Determine the Axis (ct'd)

Left Hand Lead 1	*Right Hand* Lead aVF	Quadrant	Axis
POSITIVE	POSITIVE		**Normal Axis** (0 to +90°)
POSITIVE	NEGATIVE		****Possible LAD** (0 to -90°)
NEGATIVE	POSITIVE		**RAD** (+90° to 180°)
NEGATIVE	NEGATIVE		**Extreme Axis** (-90° to 180°)



Causes of Axis Deviation

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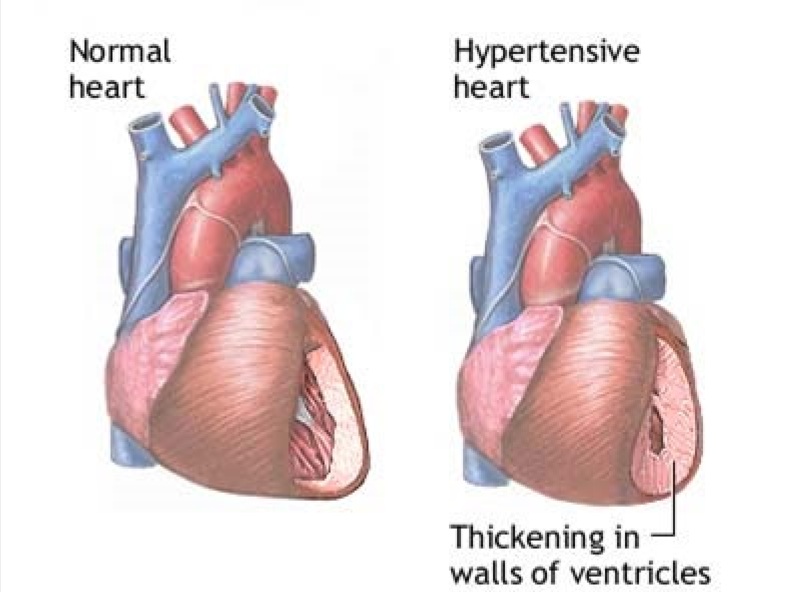
Right axis deviation:

- Right ventricular hypertrophy
- Anterolateral MI

Left Axis deviation:

- Left Ventricular hypertrophy*
- Ventricular tachycardia
- Inferior MI



Kaltura:

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No Questions related to Objective #2

Measure the PR interval

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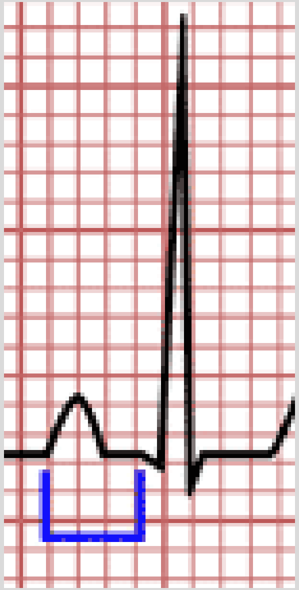
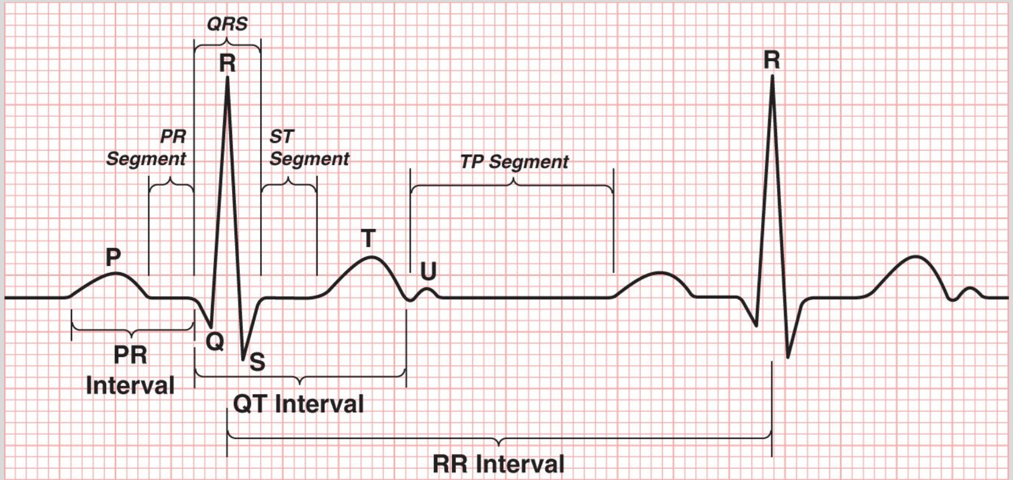
- Beginning of the P wave to the beginning of the QRS complex
- Count the number of small boxes and multiply by 0.04 seconds
  - Normal should be <200ms (5 small boxes)
- Image: PR interval = 3 small boxes X 40ms= 120ms (normal)



https://en.my-ekg.com/how-read-ekg/pr-interval.html

Measure the QRS duration

- Beginning at the start of the QRS to the end of the QRS
- Count the number of small boxes and multiple by 0.04
- Should be < 120ms
- A wide QRS can give important information on a patient's diagnosis



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Measure the QT interval

- Should be less then ½ of the R-R interval
- Indicates the time between ventricular depolarization, mechanical contraction, and ventricular repolarization
- A prolonged QT puts a patient at risk for life threatening dysrhythmias
- Numerous causes of a prolonged QT interval*
  - "ABCDEF"

*Reference Slide Notes

QT Prolongation: Drugs

Mnemonic: “Anti-ABCDE”

- Anti- **A**rrhythmic (Ia, III)
- Anti- **B**iotics (macrolides)
- Anti “**C**”ychotics (haloperidol)
- Anti **D**epressants (TCAs)
- Anti **E**metics (odansetron)
- Anti **F**ungals (azoles)

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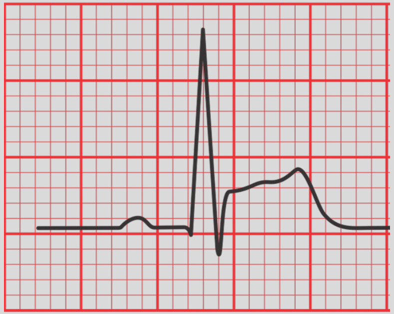
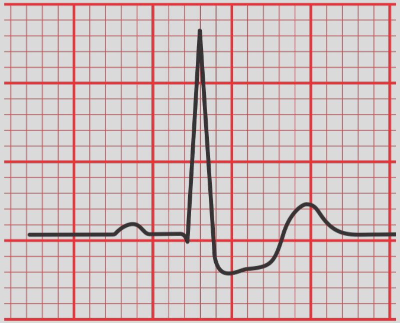
ST Segment

- ST segment should always be isoelectric
- Elevation or depression represents either ischemia or injury
- ST elevation indicates transmural ischemia
- ST depression indicates ischemia only present to the subendocardial tissue



ST elevation

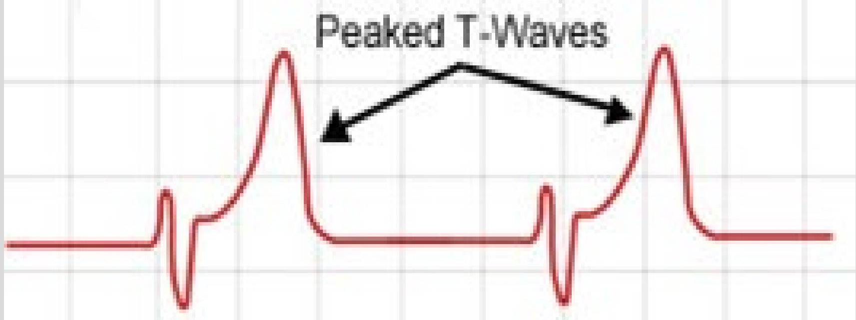


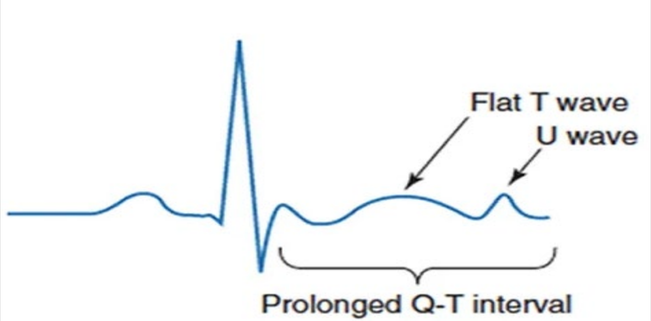
ST Depression

https://www.cvphysiology.com/CAD/CAD012

T wave

- The T wave represents ventricular repolarization (upright in lead II)
- Peaked T waves seen in hyperkalemia
- Flattened T waves and U waves seen in hypokalemia





https://coreem.net/

Putting it all together: #1

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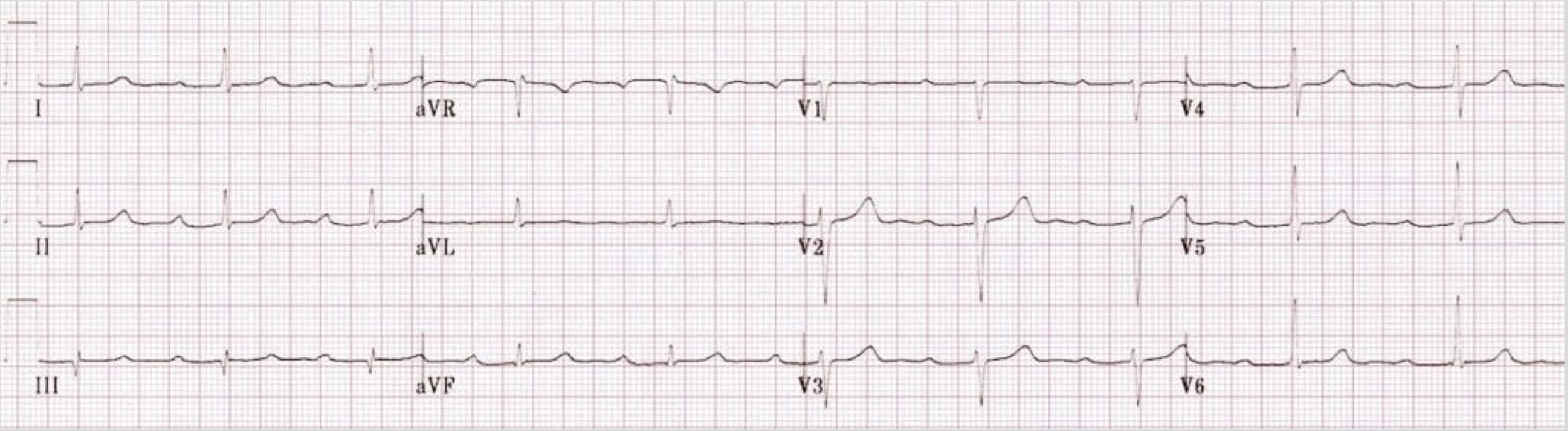

Putting it all together: #2

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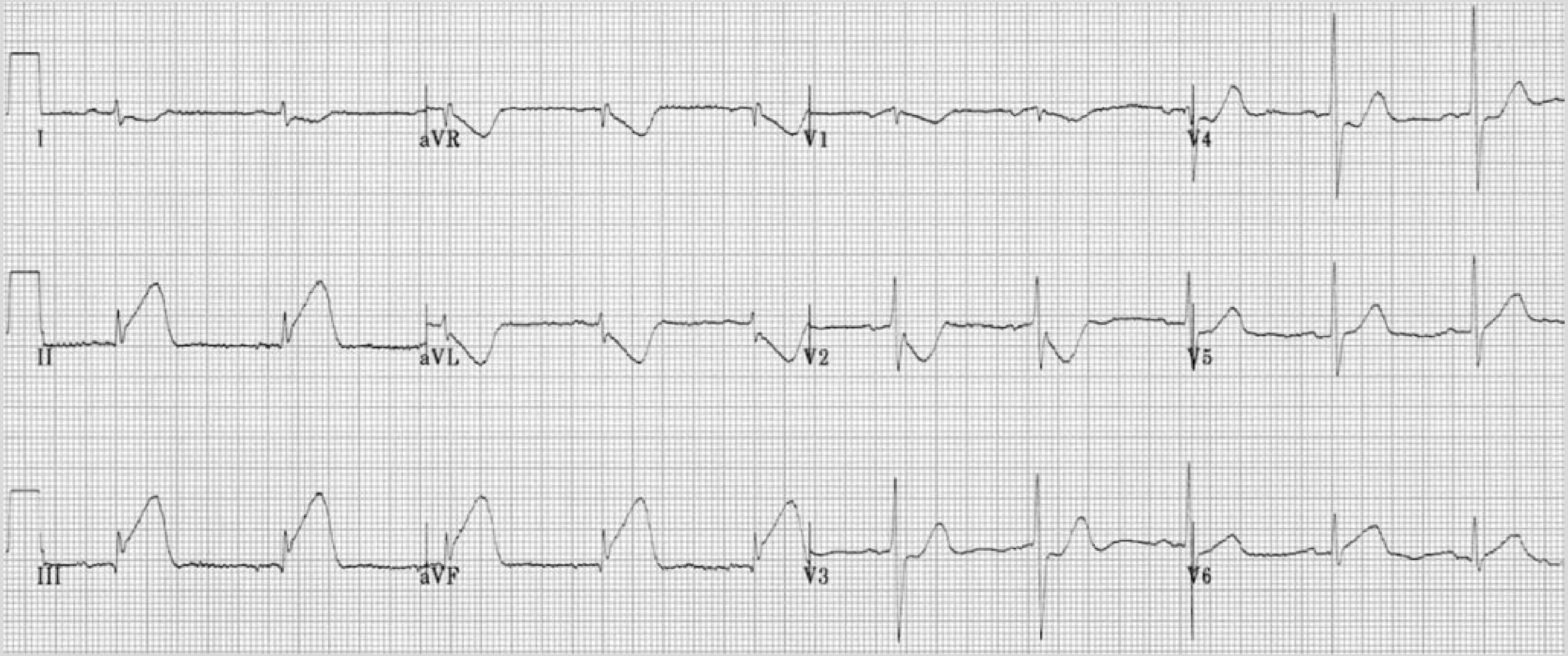

Summary Slide

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- Right Coronary Artery (RCA): II,III, aVF
- Left Anterior Descending Artery (LAD): V1-V4
- Left Circumflex Artery (LCX): V5, V6, aVL
- **Analyzing an ECG**
  1. Orient yourself
  2. Calculate Rate (three methods)
  3. Determine Rhythm (look at p waves, R-R distance)
  4. Determine the Axis (+/- in I and aVF)
  5. Calculate Intervals and QRS duration (PR interval, QT interval)
  6. Look at the ST segment (ischemia) and T wave, compare changes to the TP segment

Objective 1: Associate leads with their anatomical position and associated arteries

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- Review of coronary arteries
- Review of a STEMI; most common MI locations
- Leads correlate with specific arteries

Objective 2: Identify and interpret rate, rhythm, and axis of ECG strips

- Timing of the boxes
- Show multiple methods of finding rate
- Show rhythms - NSR, atrial fibrillation (longer method w 10 secs)
- Einthoven's triangle—> physiologic calculation of axis
- Thumbs up or thumbs down method of axis
- Causes of abnormal axis

Objective 3: Recognize normal and abnormal waves, intervals, and segments.

- Normal PR interval + abnormalities
- Normal QRS interval + abnormalities
- Normal QT interval + abnormalities
- Normal T wave + abnormalities
- Normal ST + abnormalities (STEMI)
  - transmural vs subendocardial

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Final Slide:

Questions:

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Justin Edward: jedward@nyit.edu
Grace Robinson: grobin03@nyit.edu

More exposure→ better EKG skills

-ECG Lectures **Mon 12-1 PM EST** with Dr. Cohen via Zoom -Will be sent out to the class each week by Dr. Cohen's Research Team

Feedback on the lecture:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R